

Algebra II with Trigonometry

Chapter 7.4 & 7.5

Olympiad Problems (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4626701_alg-2trig-h-chp-7-4-7-5-note-docx.pdf](#)

Problems

1. Solve in $[0, 2\pi)$:

$$\sin 5\theta + \sin \theta = \cos 4\theta + \cos 2\theta.$$

2. Find all real θ such that

$$\tan 3\theta = \tan \theta + \sec \theta.$$

3. Solve in $[0, 2\pi)$:

$$\sec \theta + \csc \theta = 4.$$

4. Find all real θ satisfying

$$\sin 4\theta \sin 2\theta = \cos 4\theta \cos 2\theta.$$

5. Solve in $[0, 2\pi)$:

$$\sin^2 \theta + \sin^2 \left(\theta + \frac{\pi}{3} \right) + \sin^2 \left(\theta + \frac{2\pi}{3} \right) = \frac{3}{2}.$$

6. Find all real θ such that

$$\cos 3\theta + \cos \theta = \sin 3\theta + \sin \theta.$$